

Q1(a)

$$\text{S.I. base Units of } k = \text{base units of } \frac{\left(\frac{Q}{t}\right)d}{A(T_2 - T_1)} = \frac{(\text{kg m s}^{-2}\text{m})}{(\text{m}^2\text{K})}$$

$$= \text{kg m s}^{-3} \text{K}^{-1}$$

Q1(b)

$$\frac{Q}{t} = kA \frac{T_2 - T_1}{d} = CT \text{ where } C = \frac{kA}{d} \text{ and } T = T_2 - T_1$$

$$\Delta T = \Delta T_2 + \Delta T_1$$

$$\frac{\Delta\left(\frac{Q}{t}\right)}{\left(\frac{Q}{t}\right)} = \frac{\Delta T}{T} = \frac{\Delta T_2 + \Delta T_1}{T_2 - T_1}$$

$$= \frac{2+2}{100-20} = \frac{4}{80} = \mathbf{0.050}$$

Q1(c)

$$\frac{Q}{t} = kA \frac{T_2 - T_1}{d} = \frac{(0.10)(100 \times 10^{-4})(100)}{2.0 \times 10^{-2}}$$

$$= 5.0 \text{ J s}^{-1}$$

$$= 5.0 \times 24 \times 60 \times 60 = \mathbf{4.3 \times 10^5 \text{ J}} \text{ in one day}$$

Q1(d)

Good electrical and thermal conductors contain many free electrons.

These free electrons are respectively capable of transporting charge and heat energy effectively .

